

Michelle Wang

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WORK EXPERIENCE

Fiserv

Berkeley Heights, NJ

Technology Analyst II, Data Science/Machine Learning

June 2025 – Present

- Design and implement a sophisticated chatbot using **Semantic Kernel** framework, incorporating Router Agent and **20+ other specialized AI Foundry agents** (plus generic plugins) along with **Azure AI Search** that chooses traditional or hybrid search based on user prompts, while leveraging **thread-based conversation histories** to preserve context regarding change managers' queries
- Build and maintain **end-to-end lifecycle APIs** for **Azure AI Foundry agents**, including agent creation, agent update, and agent deletion, while implementing **HMAC-secured authorization flows** and **role-based permissions** to enforce secure access to the chatbot and plugins associated with the agents
- Engineer and iteratively refine **ML models** using data sourced from **MongoDB** and **PostgreSQL**, working with product managers to optimize model performance and deliver insights for improving proactive risk mitigation
- Develop an automated pipeline using **Python**, **Crontab**, and **Azure Communication Services** that generates **4+ daily reports** on visualizations and **KPI** summaries covering release readiness and change-driven incidents, providing leadership with data and **LLM-based** insights and recommendations to evaluate effectiveness of processes

GMO LLC (Grantham, Mayo, Van Otterloo & Co. LLC)

Boston, MA

Research Data Analyst Co-op

August 2024 – December 2024

- Maintained robust and scalable **SQL** databases in **Microsoft SQL Server** to enhance state-of-the-art investment processes in collaboration with investment teams and external market data vendors
- Developed automated **Python** scripts for data validation, proactively identifying and resolving inconsistencies to ensure data accuracy

Lumen Technologies (formerly CenturyLink)

Remote

Internet Data Analyst Intern

May 2024 – August 2024

- Drove data-driven decision-making by comparing trends in CDR and Rate/Mbps for High Speed IP Billing Arrangements, traffic, and revenue using **4+ Tableau visualizations** (i.e., line charts, bar charts, box plots), uncovering missed high-margin customer opportunities and driving product managers to shift toward those segments

Draper Laboratory

Cambridge, MA

Machine Intelligence Co-op

July 2023 – January 2024

- Engineered **8+ ML classification algorithms** (i.e., linear discriminant analysis, neural networks) and **4+ feature selection methods** (i.e., ANOVA, lasso) using **Python** and **MATLAB** for predictive maintenance of accelerometers
- Built a **Plotly Dash** web application with **5+ interactive expenditure visualizations** and automated **Excel table** exports through a Selenium batch script, streamlining reporting workflows and reducing manual effort for Finance team

PROJECTS

ADHD Diagnostic Classification and Heart Rate Variability Analysis

- Developed **neural networks** in **R** using **Torch** package, achieving approximately **60% accuracy** in predicting ADHD given the most optimal hyperparameters
- Implemented **Autoregressive and Moving Average time series** models from scratch in **Python** to forecast heart rate variability in patients with ADHD, improving insights into physiological patterns

Emotional Intelligence Coach

- Developed an **Agentic AI** application leveraging **Ollama LLM** to enhance workplace communication through greater emotional intelligence in employees, ultimately creating value for the company

Identifying AI-Generated Text Using Machine Learning

- Preprocessed **500k essays** labeled as AI-generated or human-generated using text cleaning, lemmatization, under sampling, and TF-IDF vectorization
- Developed suite of **4+ ML models** (i.e., logistic regression, gradient boosting machines, LSTM) to predict whether an essay is AI-generated or human-generated

Sleep Efficiency Dashboard

- Developed an interactive dashboard in **Plotly Dash** integrating **Python ML algorithms** and **6+ visualizations** (i.e., density contour plot, radar chart) to help users analyze sleep patterns to improve their sleep quality

- Hypothesized that extended sleep duration, reduced caffeine and alcohol intake, non-smoking habits, and increased physical activity would collectively enhance sleep quality

Breast Cancer Survival Analysis

- Predicted breast cancer survival in years after diagnosis using patient data (273 x 1570) from Netherlands Cancer Institute using a suite of **Python ML models** (i.e., k-nearest neighbors, decision tree, random forest, PCA, multiple regression) to uncover key survival predictors (various physiological features)

Good2Go Unsupervised Learning Analysis

- Applied **K-means** clustering to uncover natural groupings among active members, visualized through **silhouette plots** and **cluster plots**
- Delivered **5+ actionable business recommendations** to improve member engagement and retention, such as targeted promotions encouraging frequent short-distance renters to book longer trips

Manhattan Rent Price Prediction

- Built explanatory models in **Python, Tableau, and Excel** to analyze key factors influencing Manhattan rent prices
- Developed robust predictive models in **Python** (i.e., KNN regressor, multiple linear regressor, random forest regressor) with $r^2 > 0.80$ to predict rent prices given the most important factors as determined by the random forest

Universal Music Database System

- Designed and implemented a robust **MySQL** database architecture, **including relational schemas, stored procedures, triggers**, to support multi-entity search, review creation/update/delete permissions, and usage analytics
- Built a **Python** application using **MySQL** connectors, delivering a seamless **CLI** experience where users are able to authenticate, manage streaming-service preferences, perform **CRUD** operations on songs/artists/albums/reviews, and retrieve personalized activity statistics

TECHNICAL SKILLS

Data Skills: Agentic AI, Data Analysis, Data Visualization, Machine Learning, Database Management, Quality Assurance

Data Tools/Languages: Python (Visual Studio, PyCharm, Jupyter Notebook), SQL (Microsoft SQL Server Management Studio, MySQL, Oracle, PostgreSQL), NoSQL (Redis, MongoDB, Neo4j), Tableau, Power BI, R, HTML, CSS, JavaScript, Postman, MATLAB, Scikit-Learn, Pandas, NumPy, Keras, Seaborn, Matplotlib, Git, Jira, Azure AI Foundry, Databricks, PowerPoint, Excel, Word

EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Data Science and Business Administration (Business Analytics)

April 2025

GPA: 3.95/4.00

Relevant Coursework: Machine Learning/Data Mining II, Large-Scale Storage, Modeling for Business Analytics, Database Design, Fundamentals of Information Analytics, Creating Business Value with Data and AI Tech, Info Viz, Prob & Stats

Activities: Club Management Lead of Khoury College Club Council, Vice President of DATA Club, Volunteer of Circle K